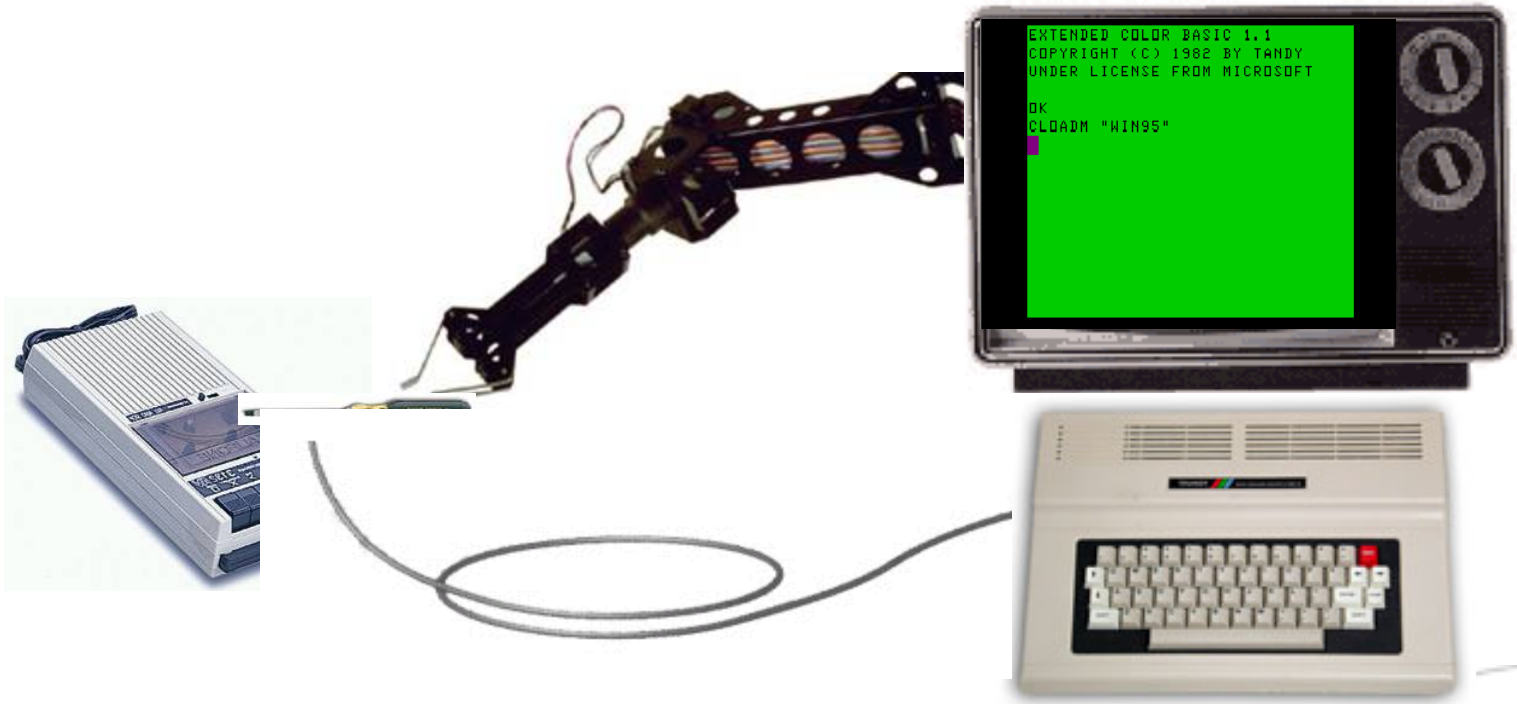
Quantum Computers That Fix Themselves



Dave Bacon
Caltech
Department of Physics
Institute for Quantum Information



The Enemy?

Caltech

The California Tech



Quantum Computing Mecca

27 Nobel Prizes

0.01 Nobel Prizes / Capita

MIT

The Tech
Established 1881

Quantum Computing Mecca

57 Nobel Prizes

0.005 Nobel Prizes / Capita

“lies, damn lies, and statistics.” – Leonard Henry Courtney

Caltech





Castor canadenis

North America Range

Beavers are found throughout North America. The only areas they are not found are the arctic tundra, most of peninsular Florida, and the southwestern desert areas. Beavers can live anywhere



The Quantum Computing Roadmap



ST Boulder Ions

1941 – First programmable electronic calculator.
mechanical relays.
1943 – ENIAC.
18000 vacuum tubes: “nature abhors the vacuum tube.”
1947 – Bell labs develops the transistor.
1952 – G. W. Dummer proposes manufacturing electronic equipment in one block with no connecting wires.
1959 – Texas Instruments and Fairchild Semiconductor invent the integrated circuit.
1964 – first IBM 360 series (the concept of an “architecture”)
1964 – Integrated circuit which cost $1000 in 1959 now costs $10. Moore describes his law.
1972 – Intel 8008 and 8 bit microprocessor.
1975 – First personal computer the Altair.
1982 – IBM PC introduced. Personal computer revolution begins.

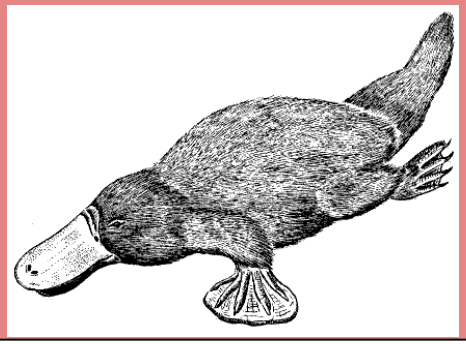
Have We Learned Anything?

History Only Teaches Me Physics

Isolated **Interacting**

What happens when many units (defined isolated) interact?

ordered
regular
reducible



turbulent
complex
not reducible

Platyputor

Computers are strange beasts in this world:

- built from parts which order and are regular
- reductionist's dream come true

yet:

- complexity of computer running a program is high
- algorithmic complexity implies reduction cannot be compressed beyond running the algorithm

Rant mode ON

The Physics Guarantee

What is the phase of matter corresponding to the computer?

PRACTICAL QUESTION (as opposed to philosophical)

There are distinct PHYSICAL reasons why robust classical computation is possible.

not all physical systems are equally good for computation:
there exists systems whose PHYSICS guarantees their ability to enact robust classical computation.

Are there (or can we engineer) physical systems whose PHYSICS guarantees robust quantum computation?

This talk: (1) this point of view for classical computers
(2) attempts to port these ideas to a quantum computer

Two Paths

Hard Drive



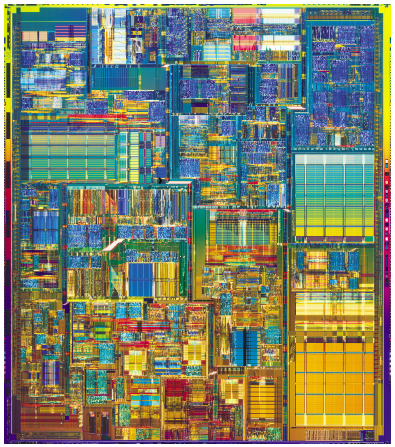
Coding: majority vote of magnetism

Error correction: local energy minimization

Fault-tolerant: macro applied field.

More than storage?

Transistor



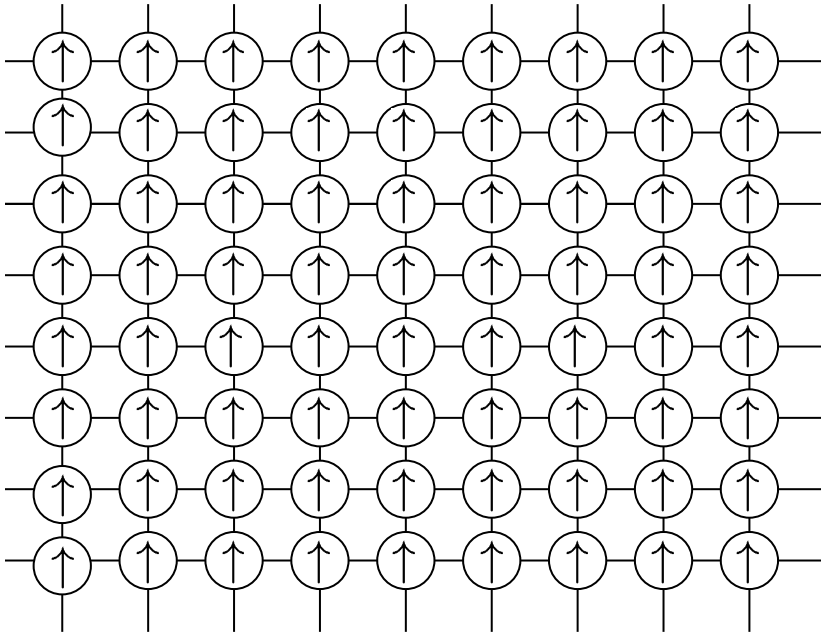
Coding: majority vote of current

Error correction: amplification

Fault-tolerant: conducting / insulating phase transition

A Dynamic Ising Model





Ising model

spins on a lattice $s_i \in \{+1, -1\}$

$$E = -J \sum_{\langle ij \rangle} s_i s_j$$

Energetically favorable to align

Order parameter:

$$M = \sum_i s_i$$

(the information)

Dynamic Ising model

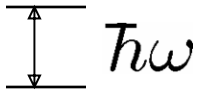
interaction with environment causes spins to flip



Reminder



two-level system radiatively coupled to a thermal reservoir:



$$\frac{\partial \boldsymbol{\rho}}{\partial t} = -\bar{n}_{th}\frac{\gamma}{2}\left[\boldsymbol{\sigma}_-\boldsymbol{\sigma}_+\boldsymbol{\rho} + \boldsymbol{\rho}\boldsymbol{\sigma}_-\boldsymbol{\sigma}_+ - 2\boldsymbol{\sigma}_+\boldsymbol{\rho}\boldsymbol{\sigma}_-\right]$$

"heating"

$$-(\bar{n}_{th}+1)\frac{\gamma}{2}\left[\boldsymbol{\sigma}_+\boldsymbol{\sigma}_-\boldsymbol{\rho} + \boldsymbol{\rho}\boldsymbol{\sigma}_+\boldsymbol{\sigma}_- - 2\boldsymbol{\sigma}_-\boldsymbol{\rho}\boldsymbol{\sigma}_+\right]$$

"cooling"

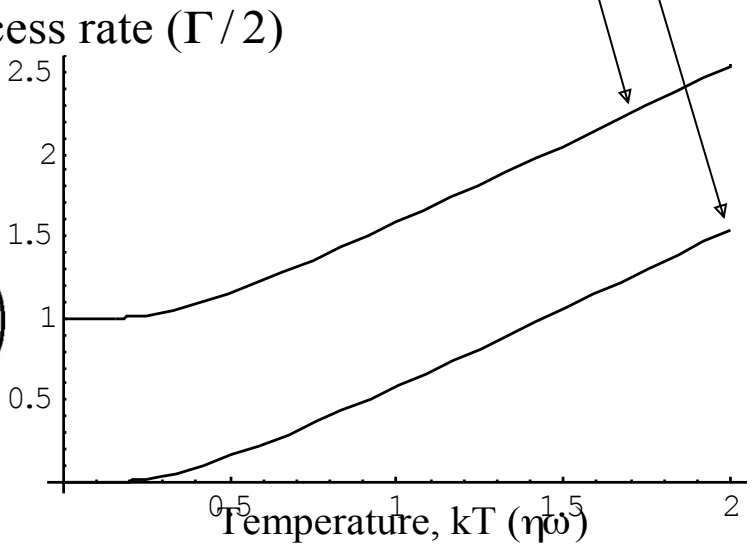
$$-\phi\left[\boldsymbol{\rho} - \boldsymbol{\sigma}_z\boldsymbol{\rho}\boldsymbol{\sigma}_z\right]$$

"non-dissipative"

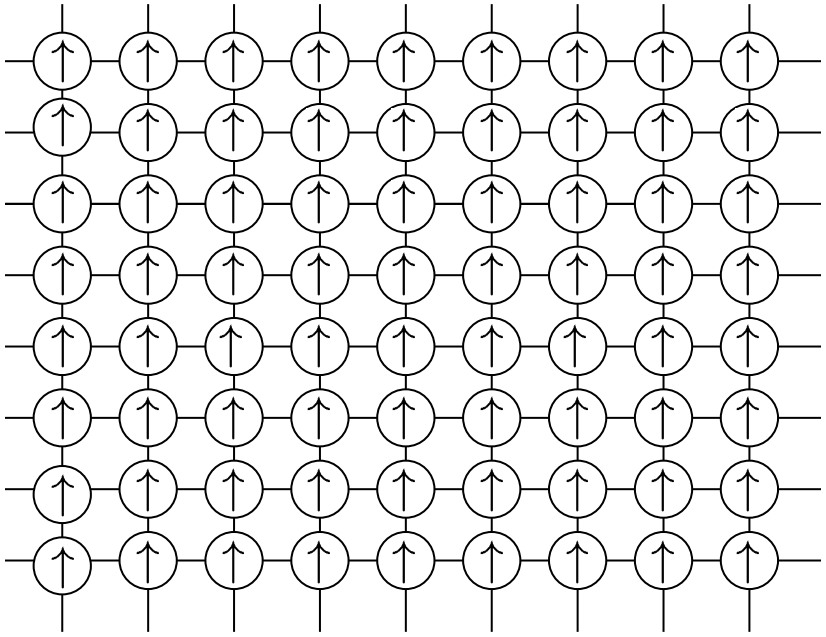
$$\bar{n}_{th} = \frac{1}{\exp\left(\frac{\hbar\omega}{kT}\right)+1}$$

$$\frac{\text{heating}}{\text{cooling}} = \frac{\bar{n}_{th}}{\bar{n}_{th}+1} = \exp\left(-\frac{\hbar\omega}{kT}\right)$$

$$= \exp\left(-\beta\Delta E\right)$$



A Dynamic Ising Model



Ising model

spins on a lattice $s_i \in \{+1, -1\}$

$$E = -J \sum_{\langle ij \rangle} s_i s_j + h \sum_i s_i$$

Energetically favorable to align

Order parameter:

$$M = \sum_i s_i$$

Dynamic Ising model (Metropolis update)

interaction with environment causes spins to flip

for every spin

if flipping spin would decrease energy, flip spin

if flipping spin would increase energy, flip spin with probability $\exp(-\beta \Delta E)$

Metropolis Monte Carlo

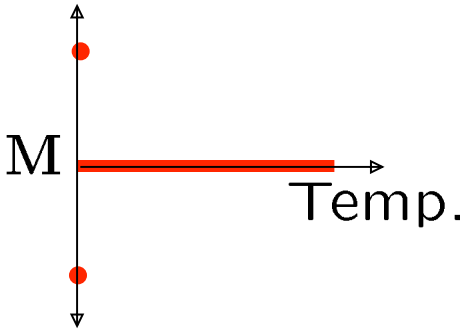
Storage and Manipulation

Compare 1D and 2D Ising models at different temperatures for
a) storage of information
b) manipulation of information

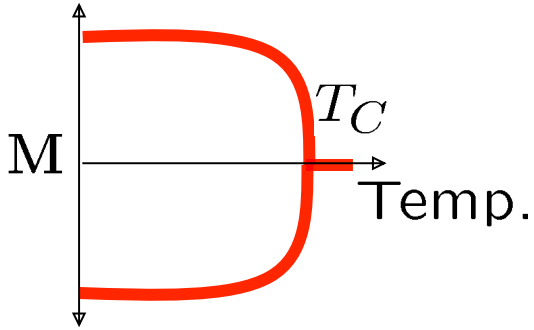
1st attempt:

Storage in thermal state (infinite time - ensemble is thermal)

1D Ising



2D Ising



Criticism: (1) thermodynamic limit taken (N infinite)
(2) what if relaxation to equilibrium takes a long time?

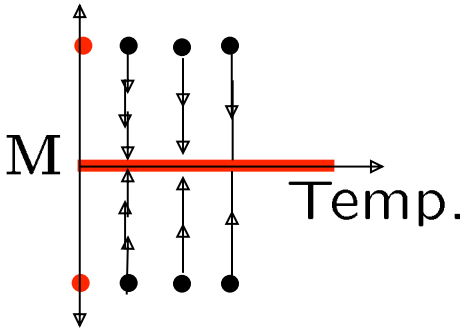
Storage



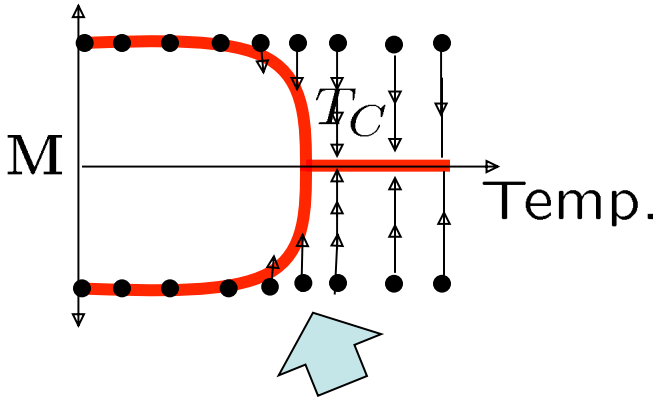
2nd attempt:

Relaxation to thermal

1D Ising



2D Ising

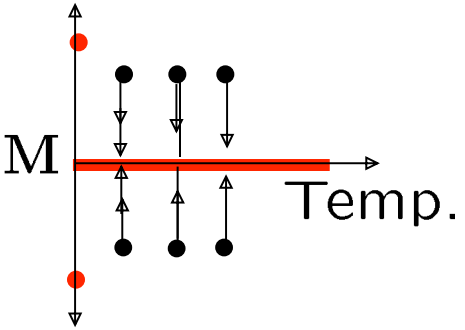


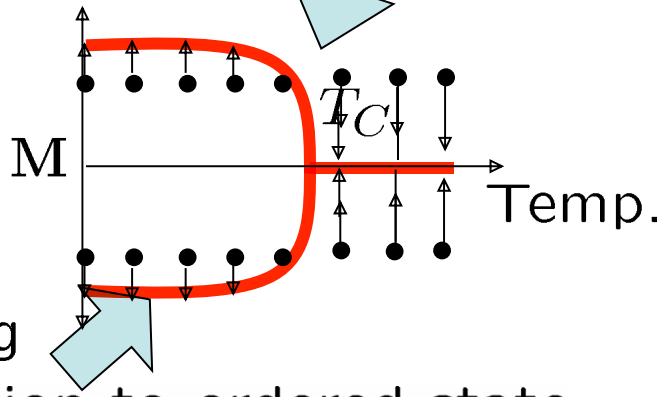
critical slowdown

$t_{th} \propto |T - T_c|^{z-\nu}$

decay rate $\propto e^{-\beta\Delta}$

Imperfect preparation





self correcting

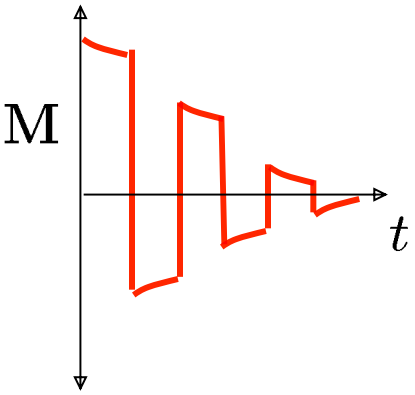
fast relaxation to ordered state

Manipulation

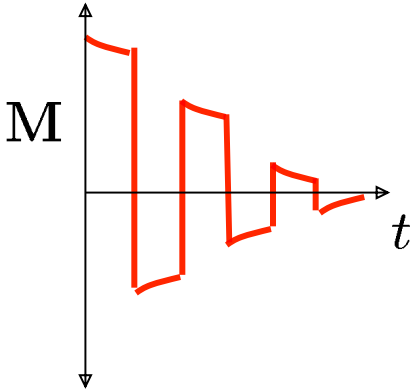


Flipping spins is imperfect

1D Ising

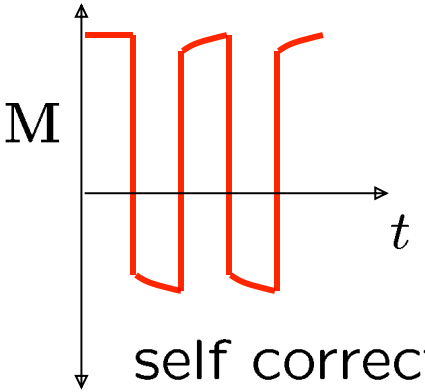


2D Ising



$T > T_C$

1D Ising
- temperature less than gap
- manipulation error must be small

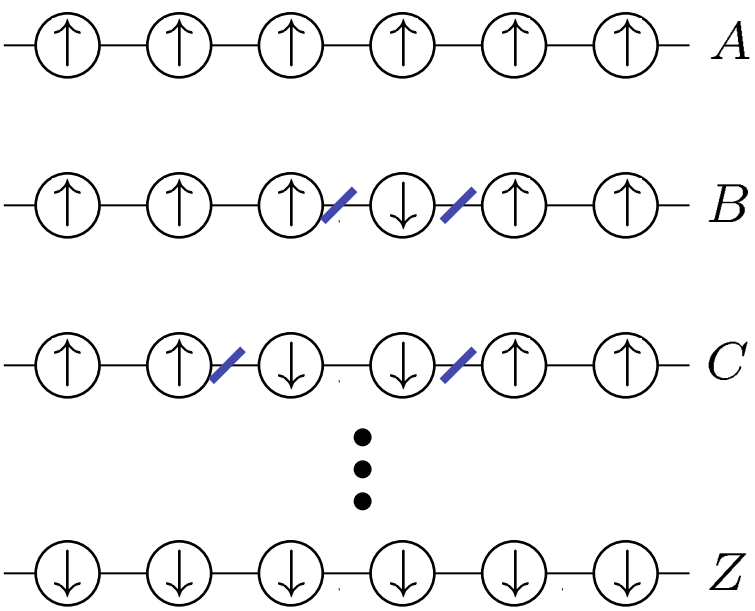


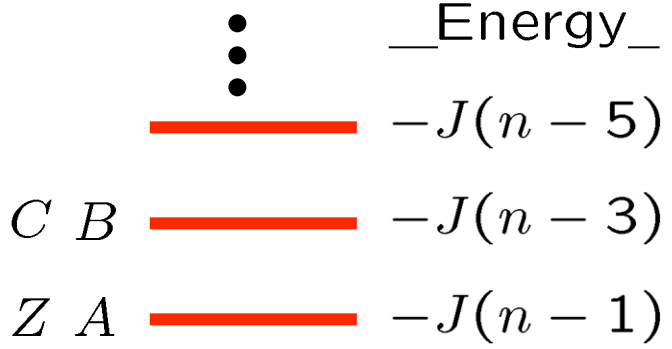
$T < T_C$

2D Ising
- temperature less than critical temp
- self correcting

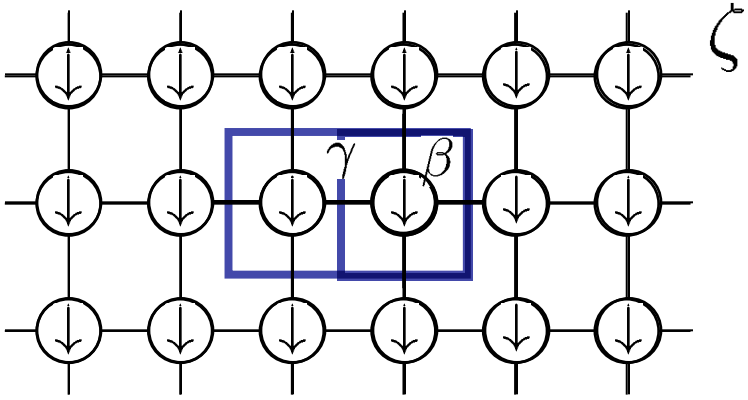
self correcting

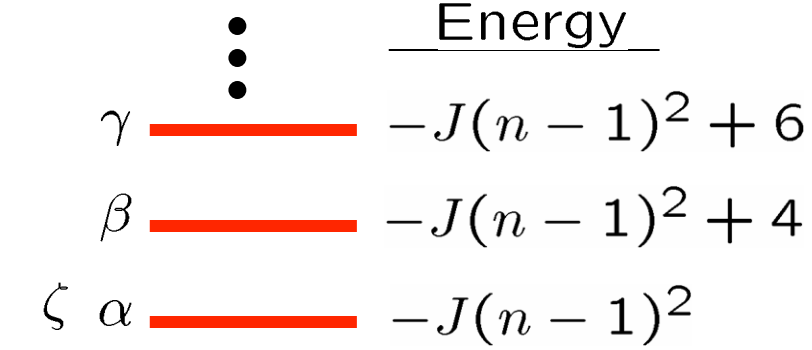
Self Fixing





Excitations free pointlike.
Energy cost constant.
disordering entropy $\gg$ energy





Excitations stringlike.
Energy cost grows like perimeter.
energy $\gg$ disordering entropy

Two Paths

Hard Drive



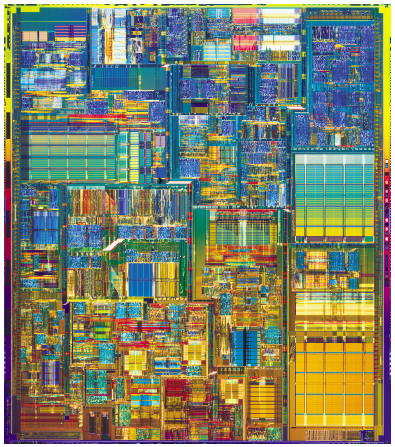
Coding: majority vote of magnetism

Error correction: local energy minimization

Fault-tolerant: macro applied field.

More than storage?

Transistor

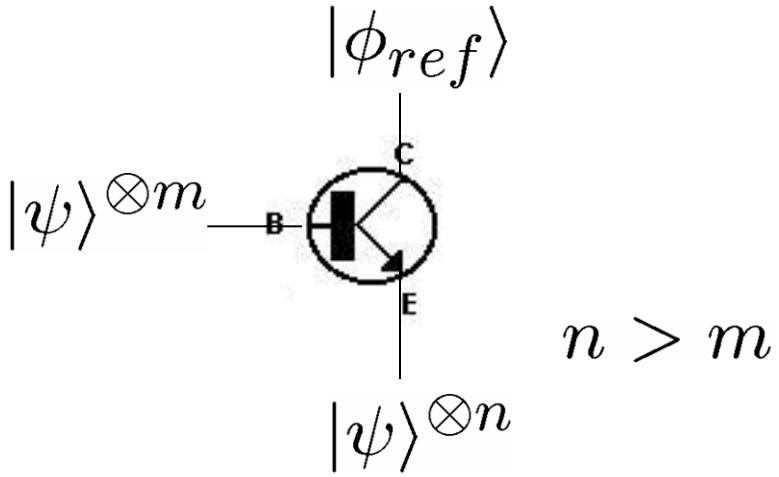


Coding: majority vote of current

Error correction: amplification

Fault-tolerant: conducting / insulating phase transition

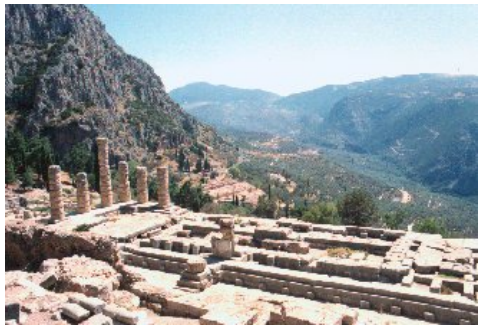
No Quantum Transistor





Quantum Cloning Machine

"A single quantum cannot be cloned," Wootters and Zurek, Nature, 1982

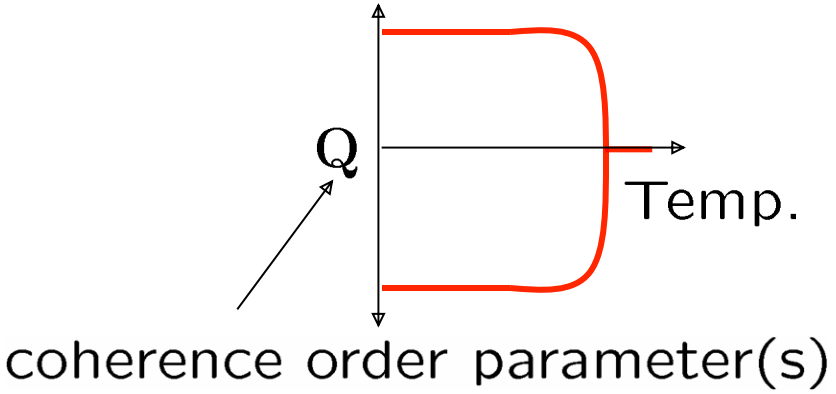


The Quantum Hard Drive?

(Kitaev)



Do there exist (or can we engineer) quantum systems whose physics guarantees fault-tolerant quantum computation?



1. Coherence preserving.

$Q_1 \Rightarrow -Q_1$

macro local operations

2. Accessible Fault-Tolerant Operations

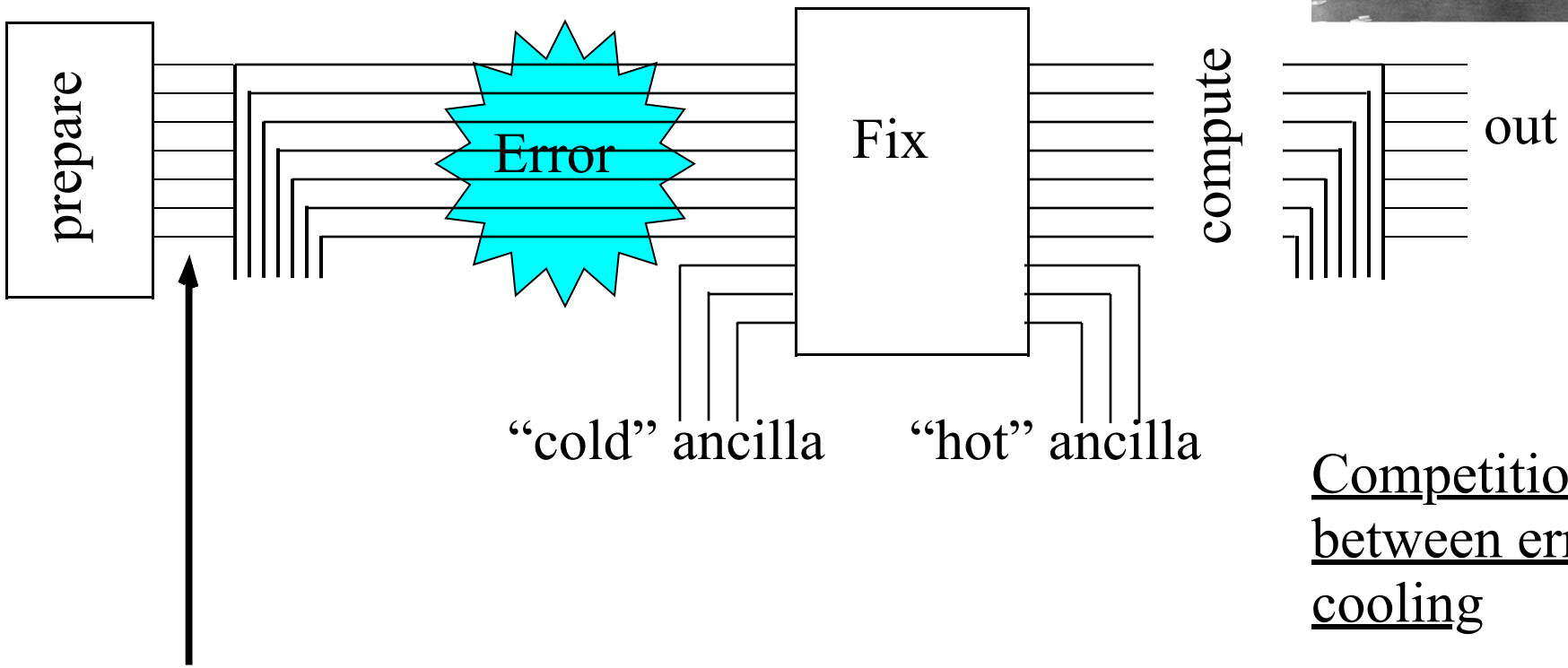
"self-correcting"

$$\left\langle G_1, \ldots, G_{n^k} \right\rangle \cong U(c^n)$$

3. Universality

Old School





Competition between errors a cooling

Quantum error correcting code

$$\mathcal{H} \subset \mathcal{V}^{\otimes n}$$

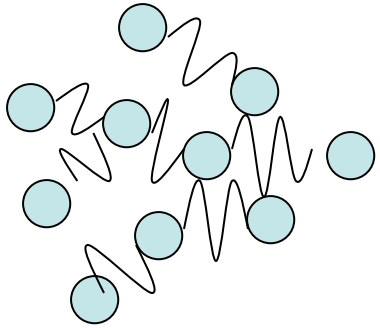
$$|i_L\rangle = \sum_\alpha c_{\alpha,i}|\alpha\rangle, \alpha \in \mathcal{V}^{\otimes n}$$

Error correcting criteria:

$$\langle i_L|\mathbf{E}_b^\dagger \mathbf{E}_a|j_L\rangle = c_{ab}\delta_{i,j}$$

If criteria holds, then fixing procedure possible

Into The Fray



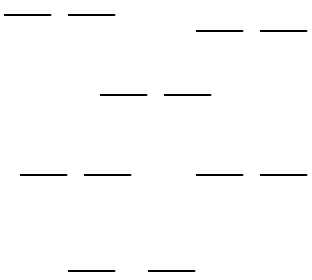
Thermodynamic limit

$$N \to \infty$$

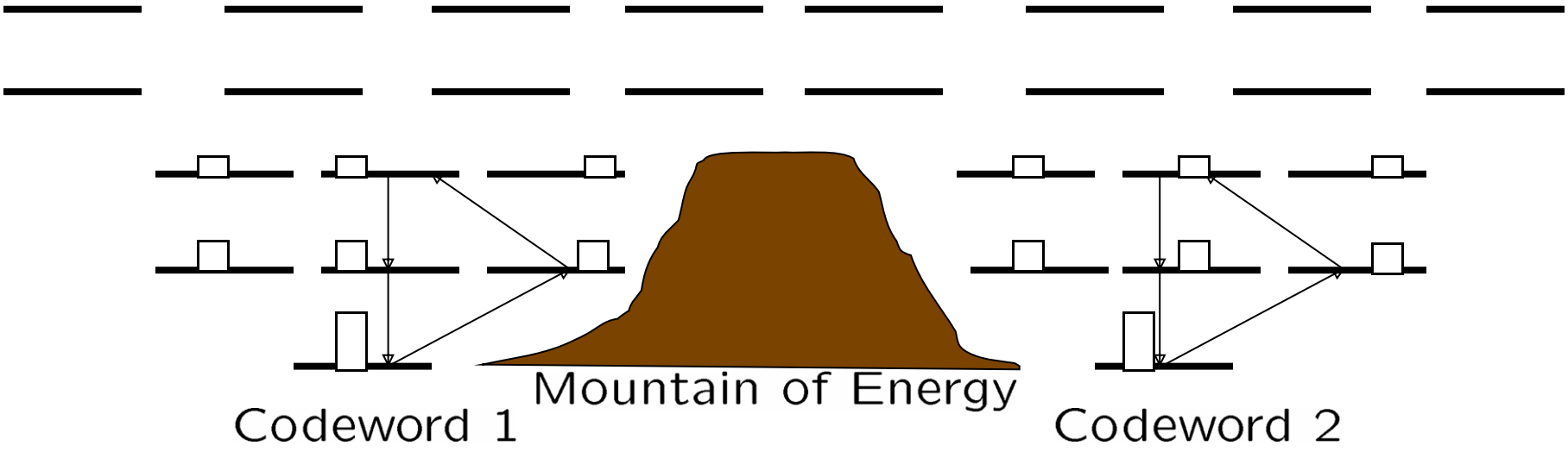
guarantees robustness

Strongly Interacting Many Body Quantum System

Hamiltonian $\mathbf{H}$

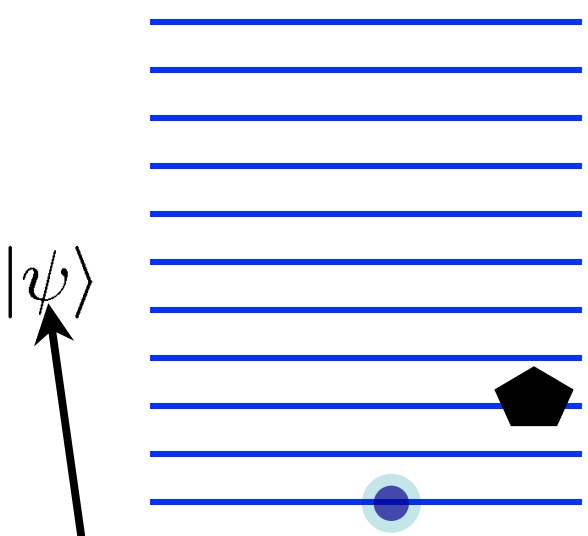


energy eigenstates

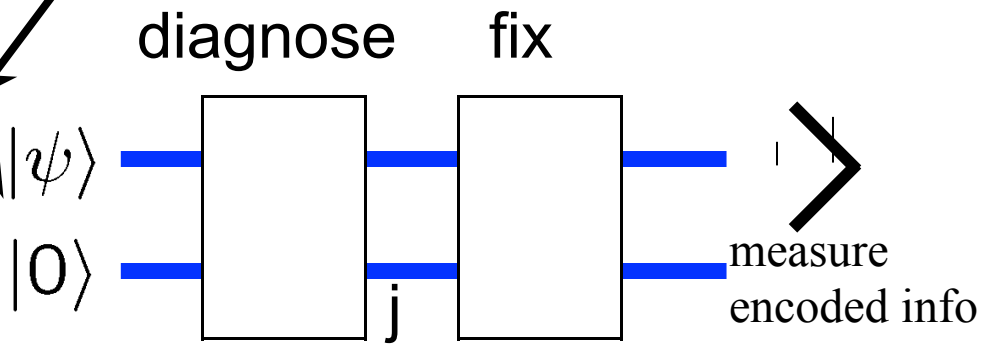


Barnes and Warren, PRL 85, 856 (2000) [but is only “classical”]

Quantum Error Correcting Order Parameters



The quantum information is still here. How do we see it?



Encoded quantum information

$$|\psi\rangle = \alpha|0_L\rangle + \beta|1_L\rangle$$

$$\mathbf{Q}_s = \sum_j \mathbf{L}_s \mathbf{F}_j \mathbf{D}_j$$

$\mathbf{L}_s$ logical encoded operation

$\mathbf{D}_j$ projector onto jth errored subspace

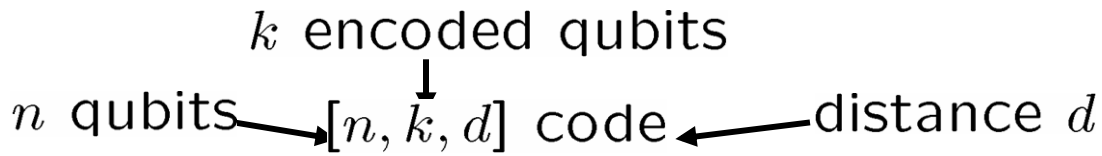
$\mathbf{F}_j$ fixing operator for jth errored subspace

A Simple Start

"To be an Error and to be Cast out is Part of [Nature's] Design."

-*William Blake*

$$\mathbf{X} = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \mathbf{Y} = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \mathbf{Z} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$



[4,2,2] quantum error detecting code:

$$\mathbf{S}_1 = \mathbf{XXXX} \quad \mathbf{S}_2 = \mathbf{ZZZZ} \qquad \mathbf{S}_i|\psi\rangle = |\psi\rangle$$

Encoded operations on encoded ($\mathbf{S}_i|\psi\rangle = |\psi\rangle$) two qubits

$$\mathbf{X}_1 = \mathbf{XXII},\ \mathbf{Z}_1 = \mathbf{IZZI}$$

$$\mathbf{X}_2 = \mathbf{IXXI},\ \mathbf{Z}_2 = \mathbf{ZZII}$$

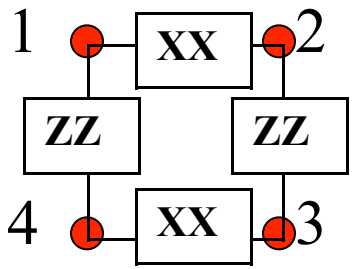
Complete basis: $|S_1, S_2, Z_1, Z_2\rangle,\ S_i, Z_i \in \{\pm 1\}$

$\mathbf{PS}_i = -\mathbf{S}_i\mathbf{P}$ for some i

$\langle i_L|\mathbf{P}|j_L\rangle = 0$, $\mathbf{P}$ single qubit operator

Supercoherence

4 qubit system



$$S_1 = XXXX$$
$$S_2 = ZZZZ$$
$$X_1 = XXII,\ Z_1 = IZZI$$
$$X_2 = IXXI,\ Z_2 = ZZII$$

System Hamiltonian: $H = XXII + IIXX + ZIIZ + IZZI$

Note symmetries: $[H, S_i] = [H, X_2] = [H, Z_2] = 0$

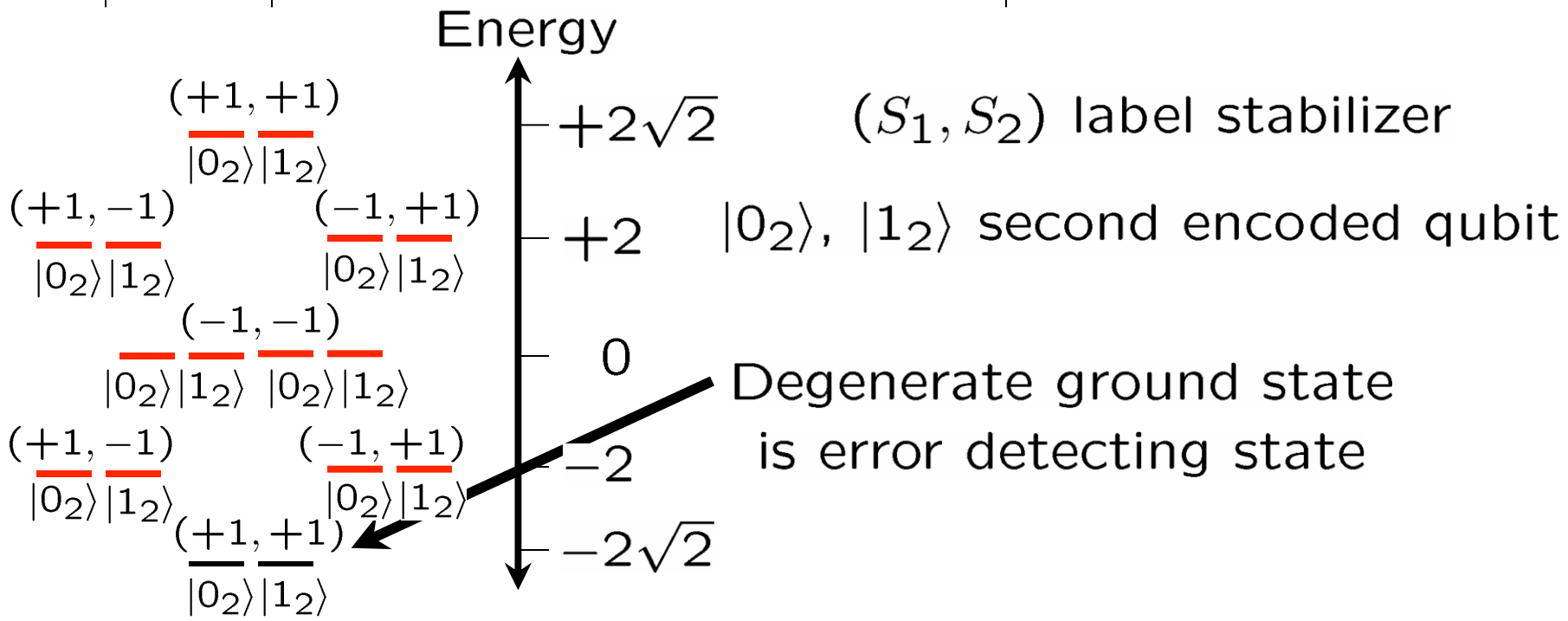
$$H = X_1(I + S_1) + Z_1(I + S_2)$$

Does not involve 2nd encoded qubit.

$\Downarrow$

every energy level at least two-fold degenerate

S_1	S_2	$\mathbf{H} = \mathbf{X}_1(1+S_1) + \mathbf{Z}_1(1+S_2)$	Eigenvalues
$+1$	$+1$	$2(\mathbf{X}_1 + \mathbf{Z}_1)$	$\pm 2\sqrt{2}$
$+1$	-1	$2\mathbf{X}_1$	± 2
-1	$+1$	$2\mathbf{Z}_1$	± 2
-1	-1	$\mathbf{0}$	0



Supercoherence

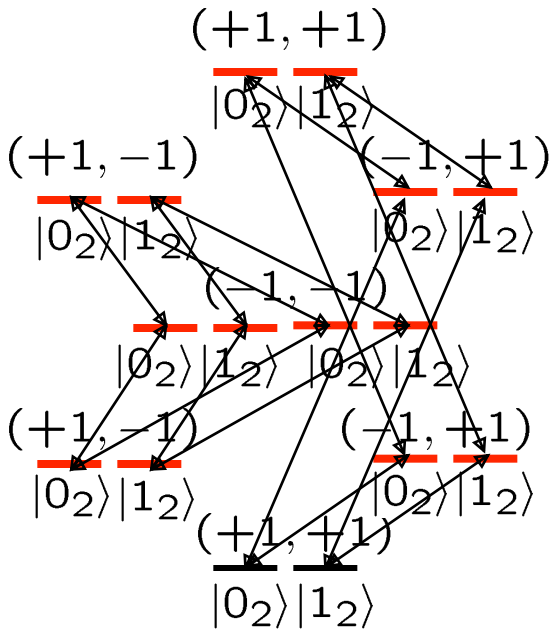
$$\mathbf{H} = \mathbf{X}_1(1 + S_1) + \mathbf{Z}_1(1 + S_2)$$

$\mathbf{S}_i\mathbf{P} = -\mathbf{P}\mathbf{S}_i$ for some i $\quad$ $\mathbf{P}$ single qubit operator

ALL single qubit errors take degenerate ground state to higher energy levels.

Single qubit errors change value of (S1,S2) and hence take ground state to higher energy level.

Effect of **ZIII**

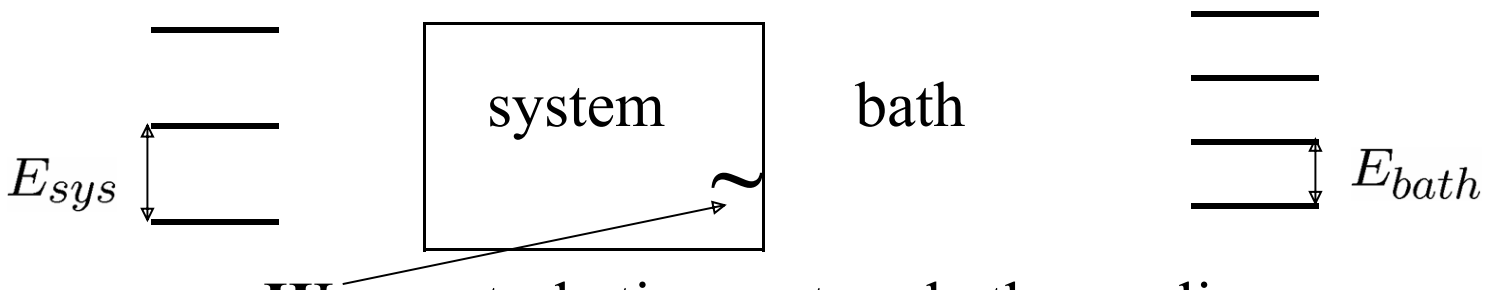


$$\underbrace{(+1, +1)}_{|0_2\rangle \qquad |1_2\rangle}$$

✗ **FORBIDDEN**

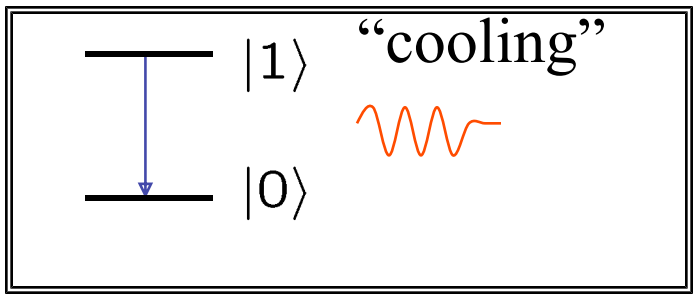
D. Bacon, Ph.D. thesis, U.C. Berkeley, 2001

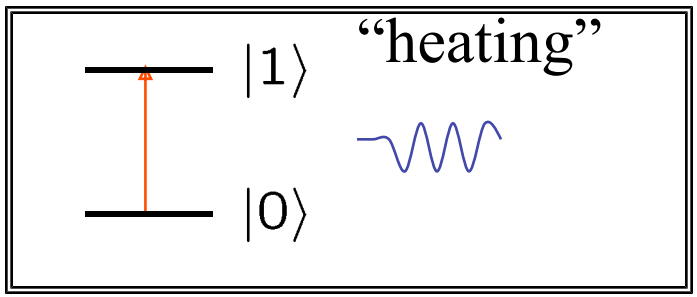
Making Decoherence Work Hard

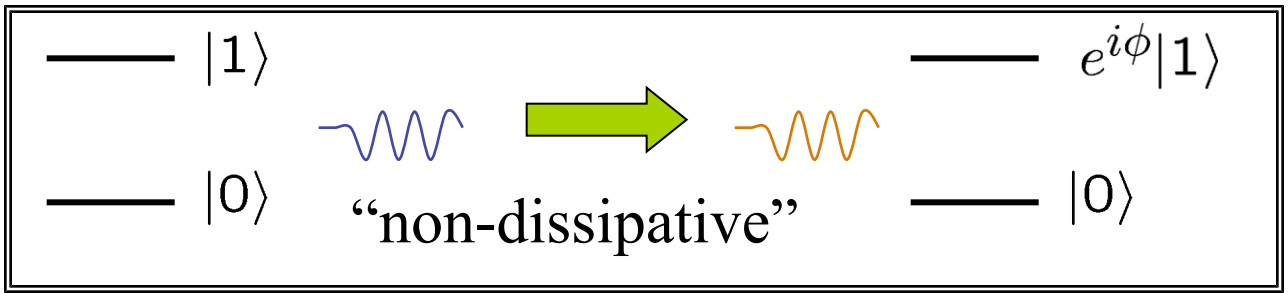


HI = perturbative system-bath coupling

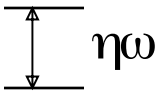
HI energy small compared to Esys and Ebath implies decoherence dominated by pathways that *conserve unperturbed energies* (essentially rotating wave approximation)







two-level atom radiatively coupled to a thermal reservoir:



$$\frac{\partial \rho}{\partial t} = -\bar{n}_{th}\frac{\gamma}{2}\left[\sigma_-\sigma_+\rho + \rho\sigma_-\sigma_+ - 2\sigma_+\rho\sigma_-\right]$$

$$-(\bar{n}_{th}+1)\frac{\gamma}{2}\left[\sigma_+\sigma_-\rho + \rho\sigma_+\sigma_- - 2\sigma_-\rho\sigma_+\right]$$

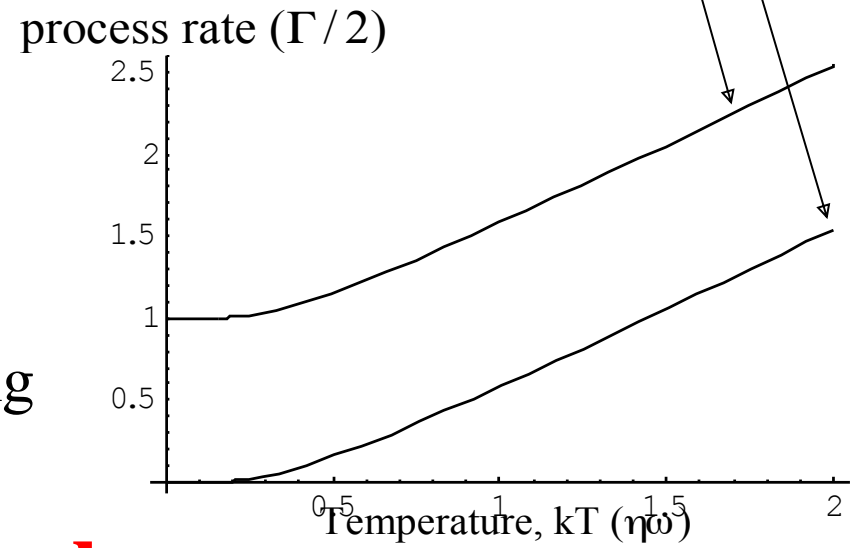
$$-\phi\left[\rho - \sigma_z\rho\sigma_z\right]$$

"heating"

"cooling"

"non-dissipative"

$$\bar{n}_{th} = \frac{1}{\exp\left(\frac{\eta\omega}{kT}\right) - 1}$$



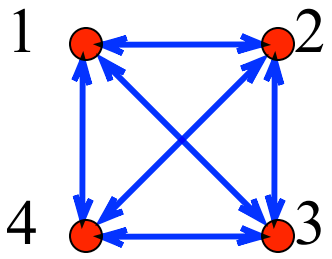
At low bath temperatures, heating disappears……..

Supercoherence made all quantum errors "heating"!

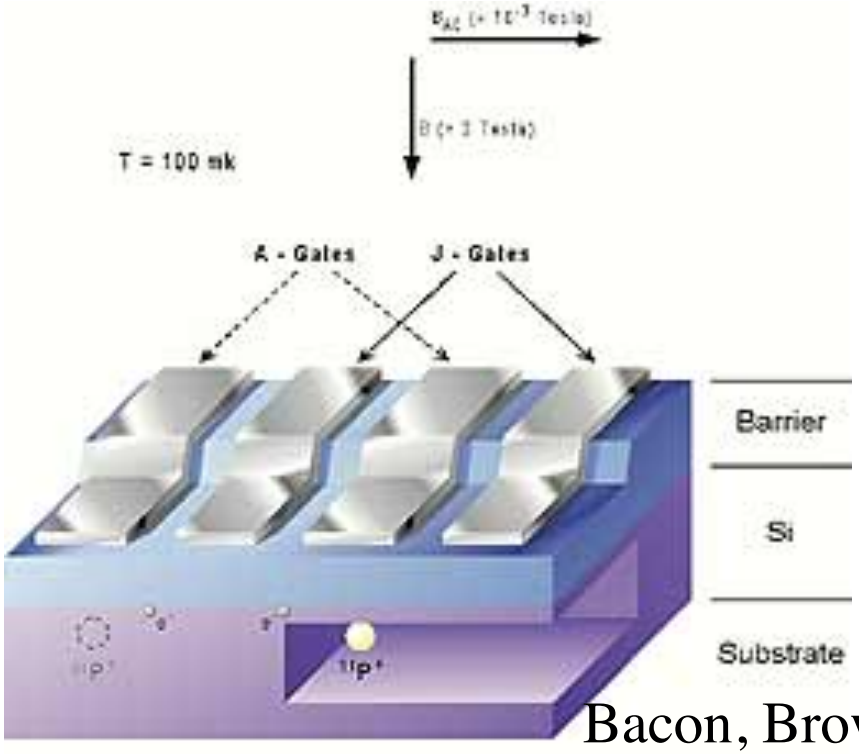
In a Few Years Time…

Four spin systems with equal strength exchange interactions between all spins:

$$\mathbf{H} = \frac{\Delta}{2}\sum_{i,j=1}^{4} \hat{\boldsymbol{s}}_i \cdot \hat{\boldsymbol{s}}_j = \frac{\Delta}{2}\hat{\mathbf{S}}^2 + c\mathbf{I}$$



Supercoherent qubit



single P atoms in Silicon

exchange controlled by voltage mediated barriers

Bacon, Brown, and Whaley, PRL **87**, 247902 (2001)

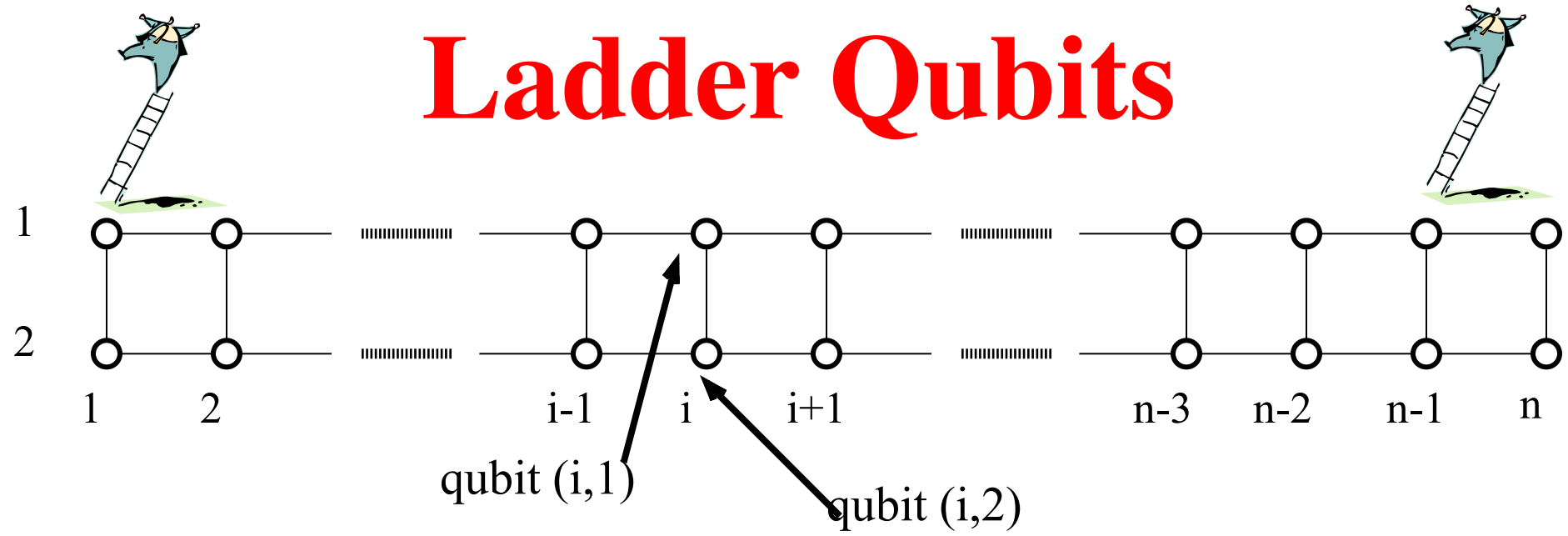
DiVincenzo, Bacon, Kempe, Burkard, and Whaley, Nature **408,** 399 (2001)

See: Hellberg, quant-ph/0304150 (2003)

Supercoherence As the First Step To Achieving the Threshold For Fault-Tolerant Quantum Computation?

Mysterious Extra Slide

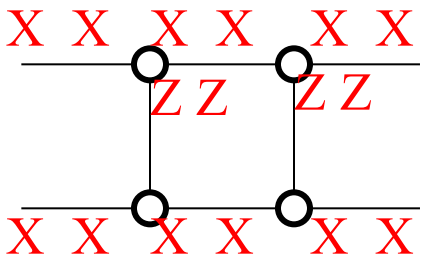
Ladder Qubits



$\sigma_\alpha^{(i,j)} \equiv \alpha^{\text{th}}$ Pauli operator acting on the (i, j) qubit tensored with I on all other qubits

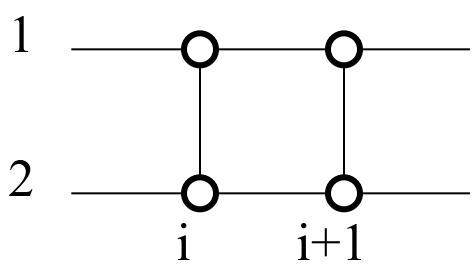
The Hamiltonian

$$H = -\omega \left[\sum_{i=1}^{n} \sigma_Z^{(i,1)} \sigma_Z^{(i,2)} + \sum_{i=1}^{n-1} \left(\sigma_X^{(i,1)} \sigma_X^{(i+1,1)} + \sigma_X^{(i,2)} \sigma_X^{(i+1,2)} \right) \right]$$



D. Bacon, Ph.D. thesis, U.C. Berkeley, 2001

Basis Change



New basis:

$$\mathbf{X}_1^{(i)} = \sigma_x^{(i,1)}, \mathbf{Z}_1^{(i)} = \sigma_z^{(i,1)}\sigma_z^{(i,2)}$$

$$\mathbf{X}_2^{(i)} = \sigma_x^{(i,1)}\sigma_x^{(i,2)}, \mathbf{Z}_2^{(i)} = \sigma_z^{(i,2)}$$

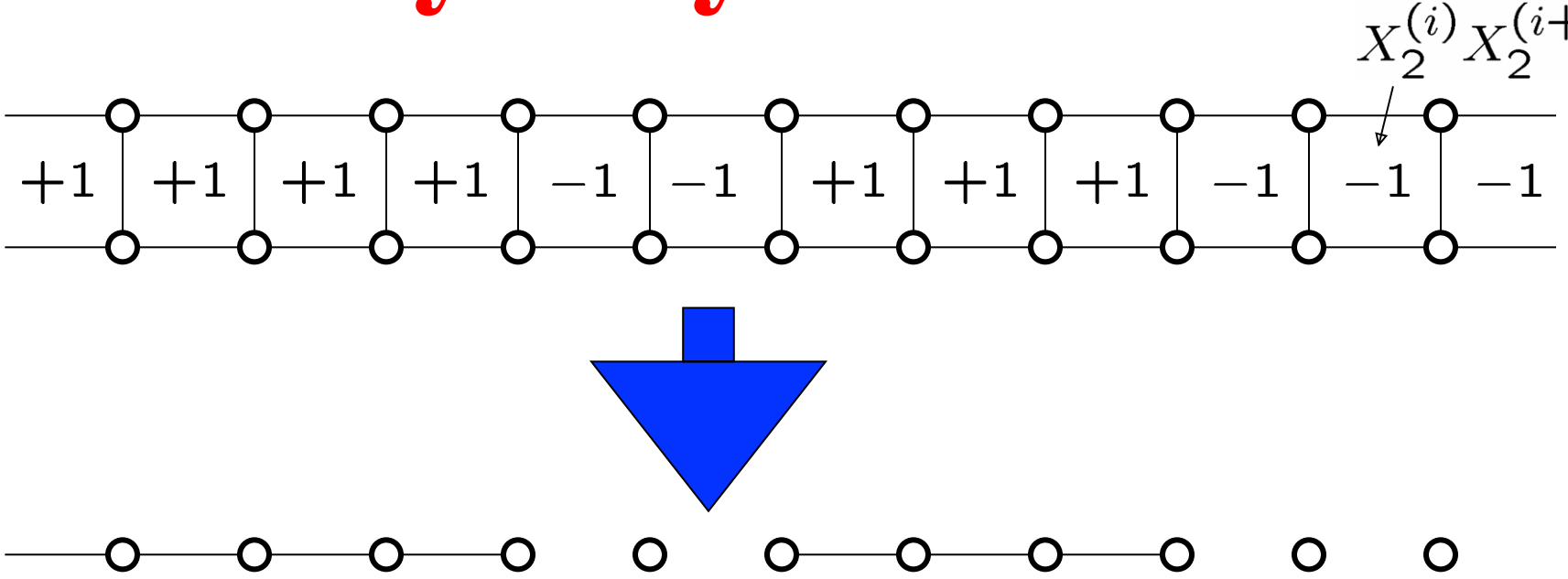
$$\mathbf{H} = -\omega\left(\sum_{i=1}^{n}\mathbf{Z}_1^{(i)} + \sum_{i=1}^{n-1}\mathbf{X}_1^{(i)}\mathbf{X}_1^{(i+1)}\left(\mathbf{I} + \mathbf{X}_2^{(i)}\mathbf{X}_2^{(i+1)}\right)\right)$$

basis $|\vec{Z}_1, \vec{X}_2\rangle = |Z_1^{(1)}, \ldots, Z_1^{(n)}, X_2^{(1)}, \ldots, X_2^{(n)}\rangle$ $\quad Z_1^{(i)}, X_2^{(i)} \in \{\pm 1\}$

$$\mathbf{H} = -\omega\left(\bigoplus_{\vec{X}_2}\left(\sum_{i=1}^{n}\mathbf{Z}_1^{(i)} + \sum_{i=1}^{n-1}c_i(\vec{X}_2)\mathbf{X}_1^{(i)}\mathbf{X}_1^{(i+1)}\right)\otimes|\vec{X}_2\rangle\langle\vec{X}_2|\right)$$

$$c_i(\vec{X}_2) = (1 + X_2^{(i)}X_2^{(i+1)})$$

Everybody's Favorite



Ising + transverse field

transverse field only

Ising + transverse field

transverse field only

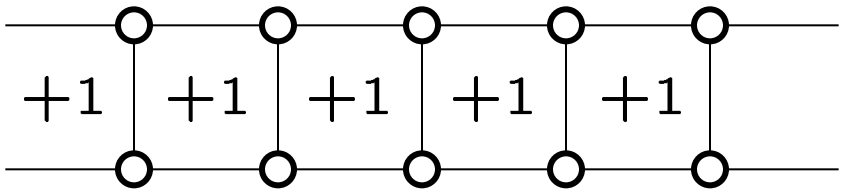
$$\mathbf{H} = -\omega \left(\sum_{i=1}^{n} \mathbf{Z}_1^{(i)} + \sum_{i=1}^{n-1} c_i(\vec{X}_2) \mathbf{X}_1^{(i)} \mathbf{X}_1^{(i+1)} \right) \quad c_i(\vec{X}_2) \in \{0, 2\}$$

Grounded



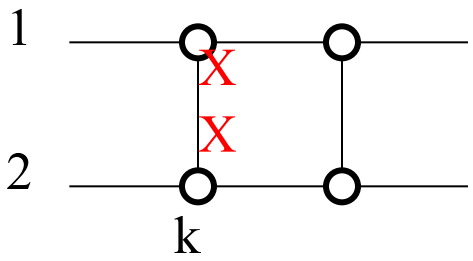
$$c_i(\vec{X}_2) = 2, \forall i$$

Ground state is two-fold degenerate



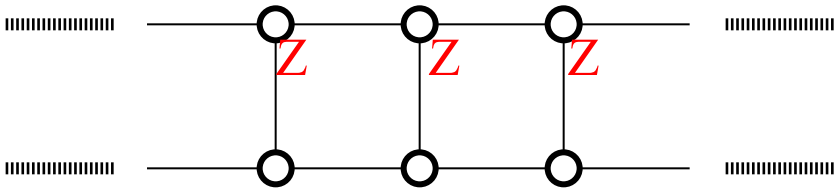
$$\mathbf{X}_2^{(i)} = +1, \forall k$$

$$\mathbf{X}_2^{(i)} = -1, \forall k$$



$$\bar{\mathbf{X}} = \mathbf{X}_2^{(k)}$$

$$[\mathbf{H}, \bar{\mathbf{X}}] = [\mathbf{H}, \bar{\mathbf{Z}}] = 0$$



$$\bar{\mathbf{Z}} = \Pi_{i=1}^{n} \mathbf{Z}_2^{(i)}$$

Detect and Correct

$$(\mathbb{C}^2)^{2n} = \mathcal{H} = \mathcal{H}_s \otimes \mathcal{H}_h \otimes \mathcal{H}_l$$

$$\mathcal{B}(\mathcal{H}) = \mathcal{B}_s \otimes \mathcal{B}_h \otimes \mathcal{B}_l$$

$\mathcal{H}_l = (\mathbb{C}^2)$ logical qubit

$\mathcal{H}_h = (\mathbb{C}^2)^n$ Hamiltonian space

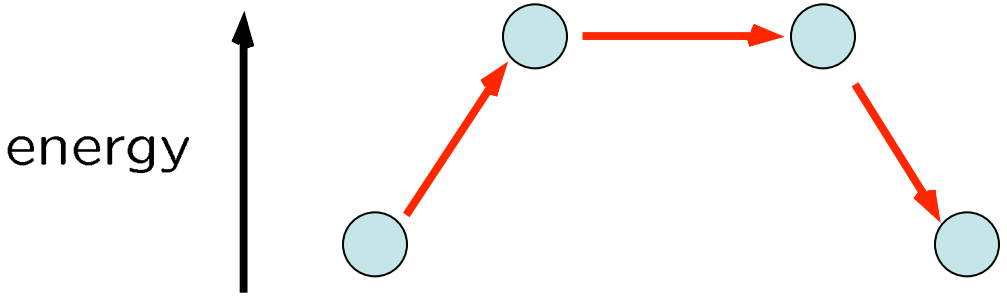
$\mathcal{H}_s = (\mathbb{C}^2)^{n-1}$ stabilizer code

$\bar{\mathbf{X}}, \bar{\mathbf{Z}} \in \mathcal{B}_l \quad \mathbf{X}_2^{(i)}\mathbf{X}_2^{(i+1)} \in \mathcal{B}_s \quad \mathbf{Z}_1^{(i)}, \mathbf{X}_1^{(i)} \in \mathcal{B}_h$

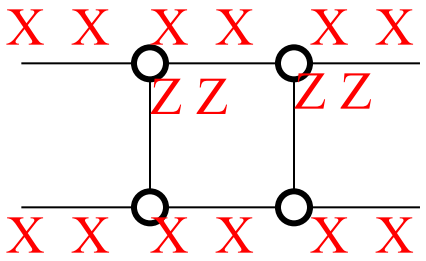
Elements of $\mathcal{B}_s$ imply distance n qubit σ_z detecting code

Can correct $\left\lfloor \frac{n-1}{2} \right\rfloor$, σ_z errors

but only 1 qubit σ_x detecting code

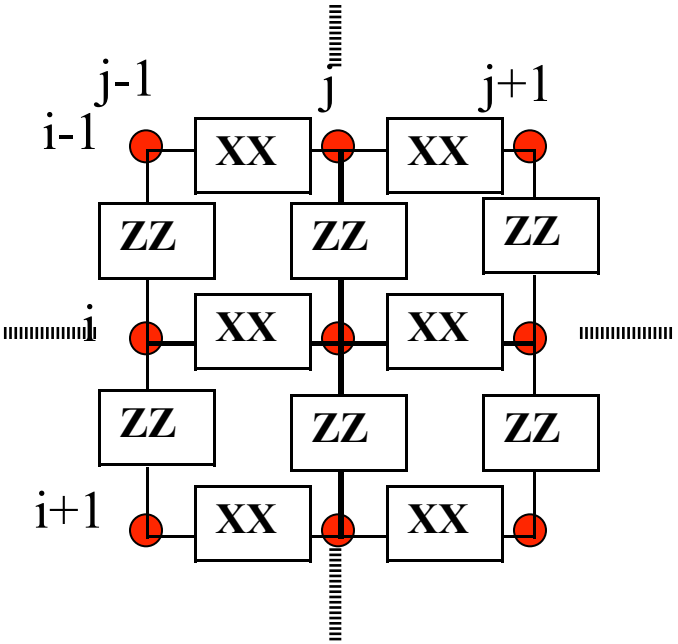


disorders at finite temperature



Into the 2nd Dimension

$$\mathbf{H} = \sum_{i=1}^{n} \sum_{j=1}^{n-1} \sigma_x^{(i,j)} \sigma_x^{(i,j+1)} + \sum_{i=1}^{n-1} \sum_{j=1}^{n} \sigma_z^{(i,j)} \sigma_z^{(i+1,j)}$$



$$[\mathbf{H}, \bar{\mathbf{X}}] = [\mathbf{H}, \bar{\mathbf{Z}}] = 0$$

Look at group algebra generated by individual bounds

Abelian invariant algebra generated double rows:

$$\mathbf{Z}_k = \prod_{j=1}^{n} \sigma_z{}^{(k,j)} \sigma_z^{(k+1,j)} \qquad \mathbf{X}_k = \prod_{i=1}^{n} \sigma_x{}^{(i,k)} \sigma_x^{(i,k+1)}$$

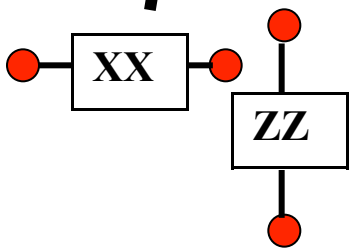
Nonabelian commutant: $\bar{\mathbf{Z}} = \prod_{j=1}^{n} \sigma_z{}^{(k,j)}$ $\bar{\mathbf{X}} = \prod_{i=1}^{n} \sigma_x{}^{(i,k)}$

Tic-Tac-"DOH!"

$$\mathcal{B}(\mathcal{H}) = \mathcal{B}_s \otimes \mathcal{B}_h \otimes \mathcal{B}_l$$

$$\mathbf{Z}_k = \prod_{j=1}^{n} {\sigma_z}^{(k,j)} {\sigma_z}^{(k+1,j)}$$

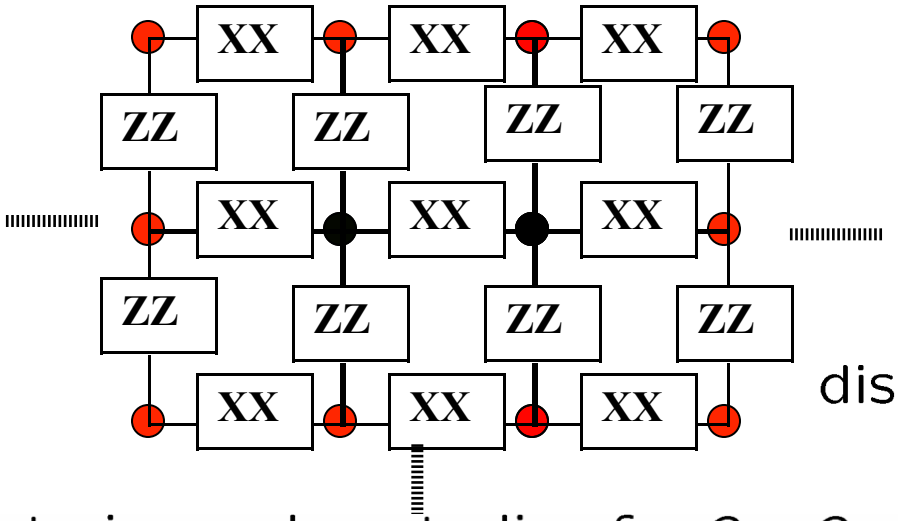
$$\mathbf{X}_k = \prod_{i=1}^{n} {\sigma_x}^{(i,k)} {\sigma_x}^{(i,k+1)}$$

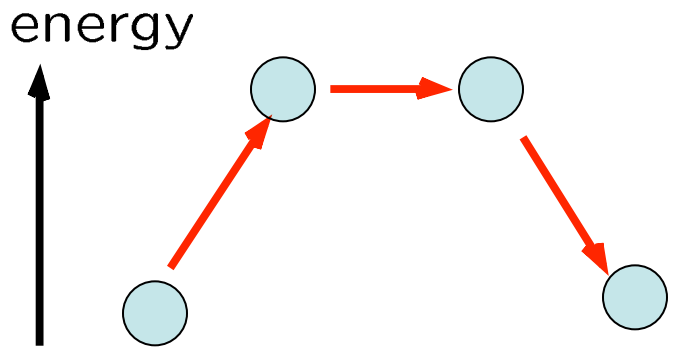


$$\bar{\mathbf{Z}} = \prod_{j=1}^{n} {\sigma_z}^{(l,j)}$$

$$\bar{\mathbf{X}} = \prod_{i=1}^{n} {\sigma_x}^{(i,l)}$$

σ_z errors

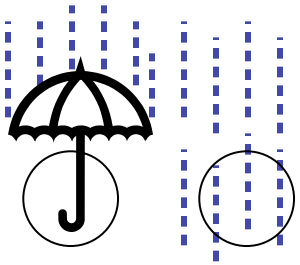




disorders at finite temperature

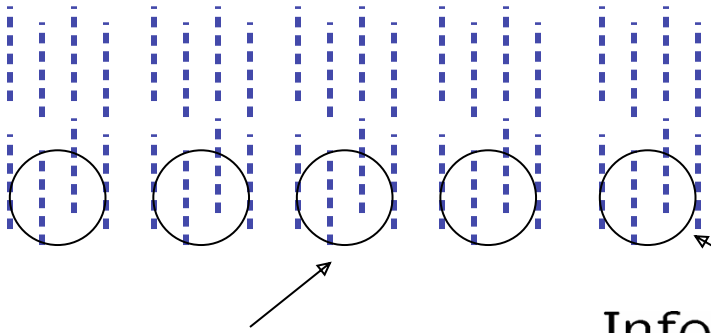
lowest eigenvalue studies for 3×3, 3×4, 4×4, 4×5 supports

Subsystem Codes



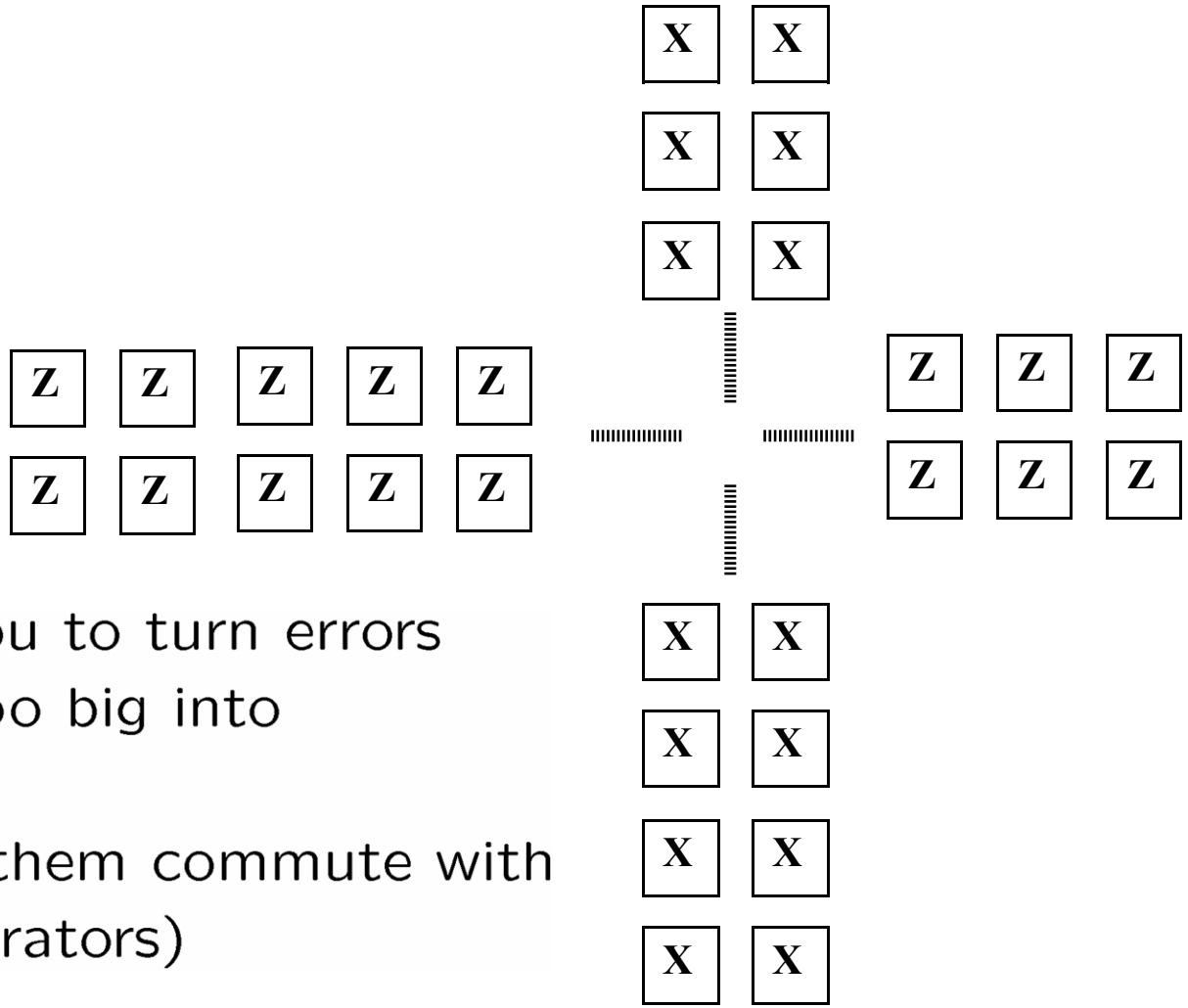
$\mathcal{H}_1 \otimes \mathcal{H}_2$

Most general notion of encoding information is into a subsystem



$$\mathbf{Z}_k = \prod_{j=1}^{n} {\sigma_z}^{(k,j)} \sigma_z^{(k+1,j)}$$

$$\mathbf{X}_k = \prod_{i=1}^{n} {\sigma_x}^{(i,k)} \sigma_x^{(i,k+1)}$$

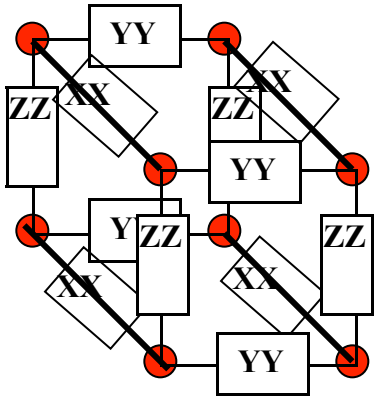


These allow you to turn errors that are not too big into elements of $\mathcal{B}_h$
(i.e. it makes them commute with the logical operators)

Into the Third Dimension

$n \times n \times n$

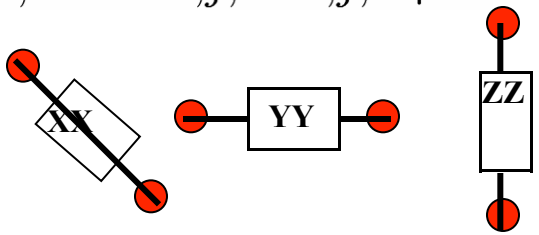
n odd



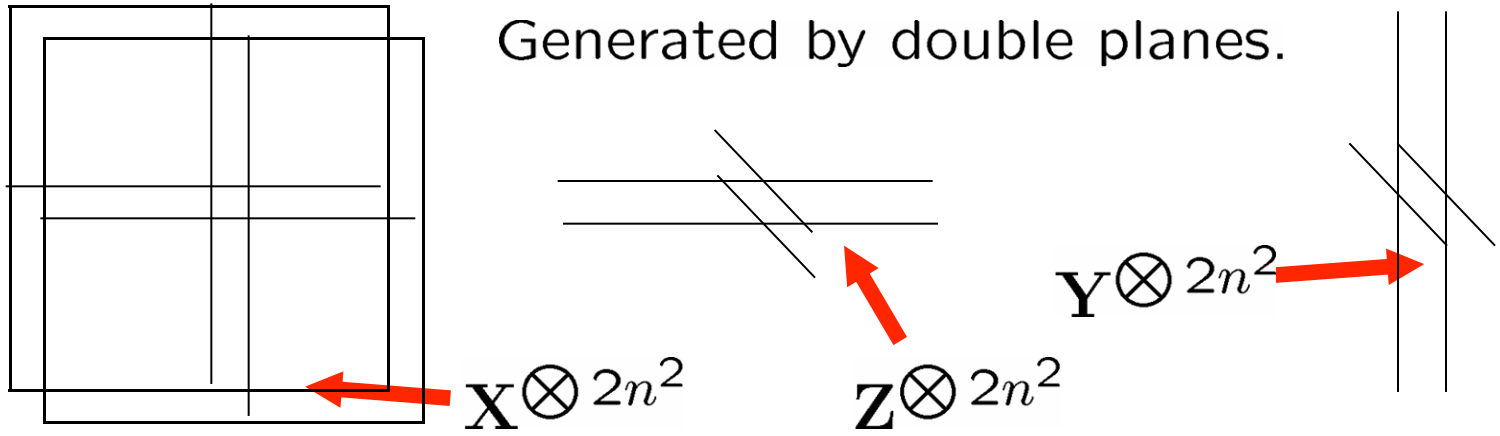
Cubic lattice with qubits at vertices

$$\mathbf{H} = -J \sum_{ijk} (\mathbf{X}_{i,j,k}\mathbf{X}_{i+1,j,k} + \mathbf{Y}_{i,j,k}\mathbf{Y}_{i,j+1,k} + \mathbf{Z}_{i,j,k}\mathbf{Z}_{i,j,k+1})$$

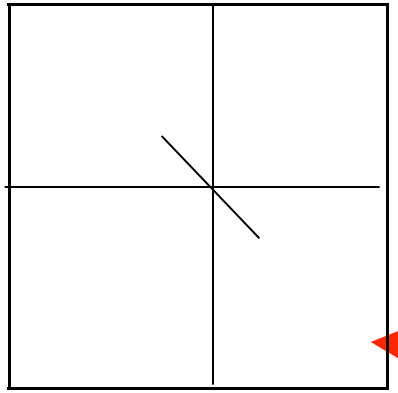
Look at group algebra generated by bonds:



Abelian invariant subalgebra:



Logical Operators



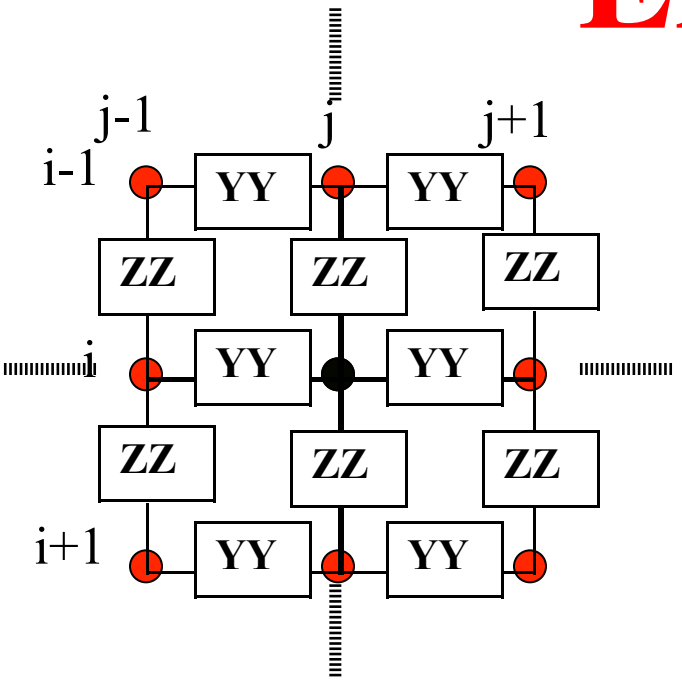
$$\bar{\mathbf{Z}} = \mathbf{Z}^{\otimes n^2}$$

$$\bar{\mathbf{X}} = \mathbf{X}^{\otimes n^2}$$

$$\bar{\mathbf{Y}} = \mathbf{Y}^{\otimes n^2}$$

$$[\bar{\mathbf{X}}, \mathbf{H}] = [\bar{\mathbf{Y}}, \mathbf{H}] = [\bar{\mathbf{Z}}, \mathbf{H}] = 0$$

Errors

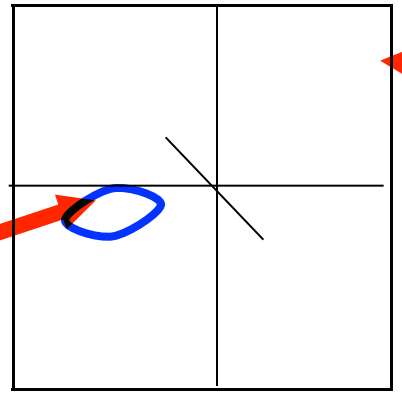


σ_z error

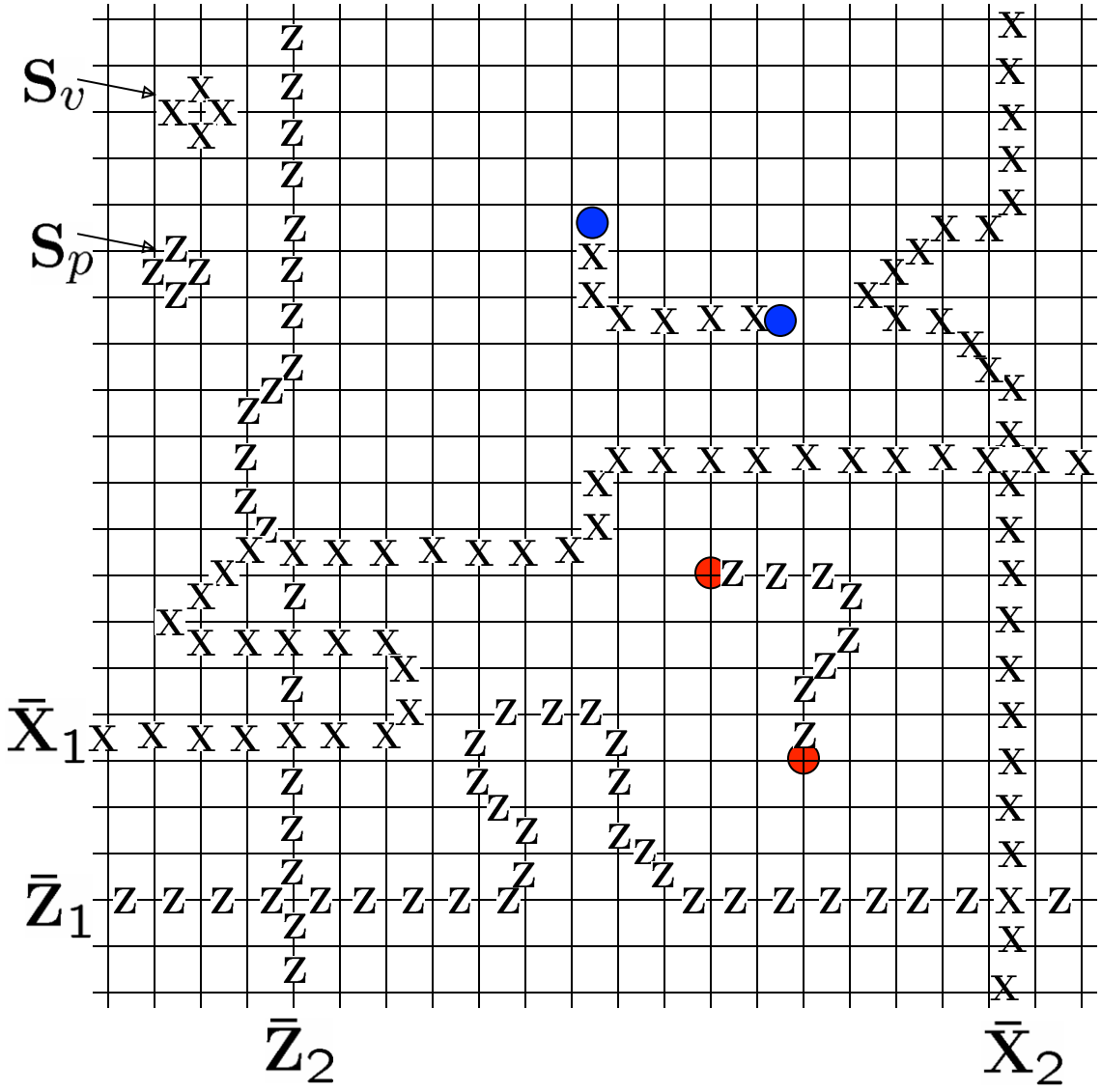
σ_y errors stringlike

σ_x errors stringlike

σ_z errors stringlike



Wherefore Art Toric Codes?



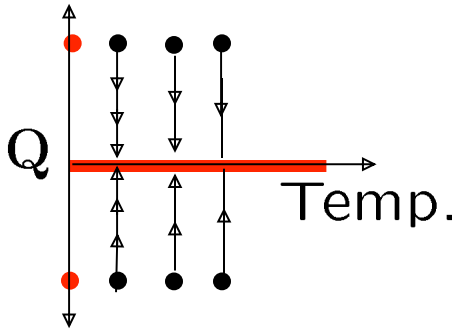
$$\mathbf{S}_p = \sum_{e \in p} \mathbf{Z}_e$$
$$\mathbf{S}_v = \sum_{e \in v} \mathbf{X}_e$$

$$\mathbf{H} = -\frac{\Delta}{2}\left(\sum_p \mathbf{S}_p + \sum_v \mathbf{S}_v\right)$$

energy gap $= \Delta$

can correct almost any $O(L^2)$ qubit errors (percolation)

natural evolution is not self correcting (we have to work in 2D)



4D Toric code should be self correcting!

Summary

Learn from our past!

Let physics do the work!

Supercoherence as first step?

Towards self-correcting quantum memories

Fin

Error correcting codes

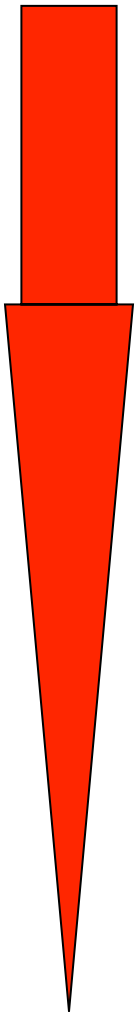
- classical
- quantum

Fault-tolerance

- local structures giving rise to global robustness (CA)

Representation theory

- identification of qubits and logical operations



Geometry, Topology?

Statistical Physics

- theory of strongly interacting quantum systems

Real Physics

- physical implementation
- optical lattices, donor electrons, superconducting circuits, etc.

Naturally Fault-Tolerant Quantum Computation